
TRAINING-FREE MONITORING OF KENYA WITH PRE-COMPUTED SATELLITE EMBEDDINGS: LAKE EXPANSION, CONSTRUCTION DATING AND CROP RETRIEVAL WITH TESSERA

A PREPRINT

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ABSTRACT

Pre-computed embeddings from geospatial foundation models turn the most expensive step of satellite analysis into shared data. We test how much analysis this enables for one country, using TESSERA, which publishes a 128-dimensional embedding for every 10 m pixel and year. An audit of Kenya shows that all 4,766 tiles centred in the country carry all nine years (2017–2025), but that the landmask removes the sea and not inland water. We then map Lake Baringo, a Rift Valley lake that has flooded its shores, by assigning each pixel to the nearer of two reference embeddings (open water and urban land), with no training. The lake grew from 181.3 km² in 2017 to 223.0 km² in 2025 (+23%), 73% of the newly flooded land went under water in 2020–2021, and our 2020 area lies 5% below an independent field-checked Landsat estimate. Classified independently each year, 67% of the changing area switches label exactly once. On the same embeddings, a single weighted change point per pixel dates 92% of changed pixels along the Nairobi–Naivasha railway to its 2018–2019 construction, while a control line opened in 2017 shows 2 changed pixels out of about 9,000; a similarity search seeded with one tea field recovers 59% of the pixels of 51 held-out tea fields with almost no detections in other landscapes; and a linear probe agrees with ESA WorldCover on 68.6% of pixels in held-out regions. Code and notebooks are released under the MIT licence.

Keywords Earth observation · geospatial foundation models · satellite embeddings · TESSERA · surface water · change detection · Kenya

1 Introduction

Most questions asked of satellite imagery are simple to state: is this pixel water, where else looks like this field, when did this hillside become a railway? Answering them is rarely simple. Optical images are hidden by cloud for weeks at a time, each place is observed on its own schedule, and the usual remedy, compositing a season into one cloud-free image, discards how the surface changes through the year, which is often the most informative part. Building a reliable per-pixel representation from raw Sentinel-1 and Sentinel-2 time series [Torres et al., 2012, Drusch et al., 2012] requires data engineering and compute that many users do not have, and in much of Africa labelled data for supervised models is also scarce.

Geospatial foundation models address this by learning general-purpose representations from unlabelled imagery [Cong et al., 2022, Tseng et al., 2023, Szwarcman et al., 2024, Clay Foundation, 2026]. A recent development is that some of them are released not as models to run but as *pre-computed embeddings*: the representation of every pixel, for every year, published as data [Brown et al., 2025, Feng et al., 2026]. If those embeddings are good enough, much of an analysis reduces to vector arithmetic.

This paper tests that proposition for one country. We use TESSERA [Feng et al., 2026], which provides a 128-dimensional annual embedding for every 10 m pixel, and ask what can be measured over Kenya with no training, or with a single linear model. Kenya is a demanding test: it spans deserts, savanna, montane forest, smallholder farmland, cities and large inland lakes, and several of its Rift Valley lakes have risen and flooded their shores during the period the embeddings cover [Tobiko, 2021].

Our contributions are:

1. **A coverage and data-quality audit** of TESSERA over Kenya (Section 5.1). Every one of the 4,766 tiles whose centre lies in Kenya has all nine annual layers (2017–2025). The landmask zero-fills the sea but not inland lakes, so “has data” must not be read as “is land”.
2. **A training-free, multi-year water map of Lake Baringo** (Section 5.2) based on the comparison of each pixel with two reference embeddings. It quantifies how much the lake grew, where and in which year the shoreline moved, and which flooded land stayed under water, and it is checked against independent estimates and for temporal consistency.
3. **Three further analyses on the same embeddings** (Section 5.3): unsupervised dating of construction, validated with a control site; one-shot retrieval of crop types, validated against held-out OpenStreetMap fields; and a linear land-cover probe evaluated with leave-one-region-out validation.
4. **Practical guidance** on pitfalls that silently corrupt results (Section 6), and an open release of all code and notebooks [Fathir, 2026].

2 Background and related work

Earth-observation foundation models. Self-supervised pre-training has produced a family of models for satellite data. SatMAE [Cong et al., 2022] adapts masked autoencoding to temporal and multispectral imagery; Presto [Tseng et al., 2023] is a lightweight transformer for pixel time series; Prithvi-EO-2.0 [Szwarcman et al., 2024] is trained on Harmonized Landsat–Sentinel-2 data at 30 m; and Clay [Clay Foundation, 2026] is an open vision transformer that accepts many sensors and resolutions. These models are usually run by the user and fine-tuned for a task.

Pre-computed embedding products. Two recent efforts publish embeddings directly. Google DeepMind’s AlphaEarth Foundations [Brown et al., 2025] provides a 64-dimensional embedding per 10 m pixel per year, derived from optical, radar, lidar and climate inputs and distributed through Google Earth Engine as the Satellite Embedding dataset [Google Earth Engine, 2026]. TESSERA [Feng et al., 2026] provides a 128-dimensional embedding per 10 m pixel per year from Sentinel-1 and Sentinel-2 time series, distributed as files under a CC0 licence [University of Cambridge Earth Observation group, 2026a,b]. Our work is a country-scale evaluation of TESSERA on tasks with independent reference data, rather than a new model.

Surface-water mapping. Open water is classically mapped with spectral indices such as NDWI [McFeeters, 1996] and MNDWI [Xu, 2006], and at global scale from the Landsat archive [Pekel et al., 2016]. Such methods typically require cloud screening, compositing and per-scene thresholds. Our water rule instead compares annual embeddings with two fixed reference vectors and needs no threshold.

Change detection in time series. Methods such as BFAST [Verbesselt et al., 2010] and LandTrendr [Kennedy et al., 2010] segment spectral time series to detect and date disturbance; a broader review of offline change-point methods is given by Truong et al. [2020]. We apply the simplest member of this family, a single weighted change point, directly to sequences of embeddings.

Spatial validation. Pixels close together are strongly correlated, so random train/test splits overstate how well a model transfers to new places [Roberts et al., 2017, Ploton et al., 2020]. We therefore evaluate the land-cover probe by holding out whole regions.

3 Data

TESSERA embeddings. TESSERA [Feng et al., 2026] processes the Sentinel-2 optical time series (10 spectral bands) and the Sentinel-1 radar time series of each pixel over a year with two transformer encoders, one per sensor, and fuses their outputs with a small network into a 128-dimensional vector. It is trained with the Barlow Twins objective [Zbontar et al., 2021], without labels, so that its output is stable across different random subsets of a pixel’s observations.

The released product is organised in tiles of $0.1^\circ \times 0.1^\circ$ (about 11×11 km), each a $H \times W \times 128$ array in a UTM projection, quantised to int8 with accompanying float32 scale factors. One tile-year occupies about 163 MB. We access dataset version 1 through the `geotessera` library (version 0.9.0) [University of Cambridge Earth Observation group, 2026a] and pin the version throughout, because embeddings from different dataset versions are not comparable.

Reference and auxiliary data. Country and province boundaries and lake outlines are from Natural Earth at 1:10m [Natural Earth, 2026]. Railway alignments, the Konza Techno City plot and crop-field outlines are from OpenStreetMap [Haklay and Weber, 2008]. Land-cover labels are from ESA WorldCover 2021 v200 [Zanaga et al., 2022], a 10 m product with a reported global overall accuracy of 76.7% [ESA WorldCover consortium, 2022]. Construction dates are from public project records [AidData, 2026, Wikipedia contributors, 2026a,b], and published lake-area estimates from Maina et al. [2026] and Tobiko [2021]. None of these sources was used to build the embeddings, so they serve as independent answer keys.

4 Methods

4.1 Notation and preprocessing

Let $\mathbf{e}_{p,t} \in \mathbb{R}^{128}$ be the embedding of pixel p in year t . We write $\hat{\mathbf{e}} = \mathbf{e}/\|\mathbf{e}\|_2$ for the unit vector, so the dot product of two unit vectors is their cosine similarity. Pixels outside the TESSERA landmask carry a vector of exact zeros rather than missing values; we define a pixel as *valid* if its embedding is finite and non-zero, leave invalid vectors at zero when normalising, and exclude them from every statistic.

4.2 Reference embeddings and the water rule

A reference embedding summarises a tile whose identity is not in doubt. For a tile $\mathcal{T}$ we average unit vectors, so that a few high-magnitude pixels cannot dominate, and renormalise:

$$\mathbf{r}_{\mathcal{T}} = \frac{\sum_{p \in \mathcal{T}} \hat{\mathbf{e}}_p}{\left\| \sum_{p \in \mathcal{T}} \hat{\mathbf{e}}_p \right\|_2}. \quad (1)$$

We build a water reference $\mathbf{r}_w$ from a tile of open Lake Victoria (centre 33.95°E , 1.05°S) and a land reference $\mathbf{r}_\ell$ from a tile of central Nairobi (36.85°E , 1.25°S), both from 2024. A valid pixel is labelled water when it is closer to the water reference:

$$\text{water}(p, t) = \mathbf{1}[\hat{\mathbf{e}}_{p,t}^\top \mathbf{r}_w > \hat{\mathbf{e}}_{p,t}^\top \mathbf{r}_\ell]. \quad (2)$$

The same two references are used for every year, so year-to-year differences come from the ground and not from the yardstick. The rule has no threshold and no training data.

4.3 Lake extent, flooding and stability

Lake Baringo (0.63°N , 36.06°E) lies inside a 3×3 block of tiles spanning $35.9\text{--}36.2^\circ\text{E}$ and $0.5\text{--}0.8^\circ\text{N}$. The block straddles the boundary between UTM zones 36 and 37, so for each year we apply Equation 2 to each tile in its native projection and resample the labels with nearest-neighbour interpolation onto one shared 10 m grid in UTM zone 37N (EPSG:32637). On this grid each pixel covers exactly 100 m^2 , so areas are pixel counts. The lake in year t is the largest connected component of water pixels; other water (rivers, ponds) is reported separately. We check that the lake never touches the edge of the block.

For every pixel that was land in 2017 and lake in a later year we record the *first year under water*. We separate pixels that stayed under water in every year from their first flooding through 2025 from those that dried out again, and we record lake pixels from 2017 that became land. Because each year is classified independently with no temporal smoothing, the number of lake/land switches per pixel over the nine years is a direct test of noise: a moving shoreline produces few, consistent switches, while classification noise produces flicker.

4.4 Dating change with a single change point

For each pixel with valid embeddings in all $T = 9$ years, and for each split $b \in \{1, \dots, T - 1\}$, we compare the mean direction of the years before and after the split:

$$d_b = 1 - \mathbf{a}_b^\top \mathbf{z}_b, \quad \mathbf{a}_b = \frac{\sum_{t=1}^b \hat{\mathbf{e}}_t}{\left\| \sum_{t=1}^b \hat{\mathbf{e}}_t \right\|_2}, \quad \mathbf{z}_b = \frac{\sum_{t=b+1}^T \hat{\mathbf{e}}_t}{\left\| \sum_{t=b+1}^T \hat{\mathbf{e}}_t \right\|_2}. \quad (3)$$

Table 1: Sites used to evaluate change dating, with their published construction dates.

Site	Change	Published dates
SGR Nairobi–Naivasha (Phase 2A)	120 km of new railway down the Rift escarpment	main works Jan 2018–Sep 2019 [AidData, 2026]
SGR Mombasa–Nairobi (control)	first standard-gauge line	opened 31 May 2017 [Wikipedia contributors, 2026a]
Konza Techno City	new city on open plain	Phase 1 roads (170 ha) finished 2022 [Wikipedia contributors, 2026b]

The split is chosen as $b^* = \arg \max_b d_b \sqrt{b(T-b)/(T/2)}$; the weight is the usual change-point weighting that keeps a single anomalous first or last year from appearing as a lasting change [Truong et al., 2020]. Each pixel receives a change magnitude $D = d_{b^*}$ and a date, the first year of the new state. Every pixel receives a date, so a date is only meaningful where D is large. We define “changed” as D above the 99th percentile of D over background pixels, set before inspecting the target sites.

We evaluate the method at two railway lines and one new city, all with published dates (Table 1), on seven tiles (63 tile-years). Zones are rasterised from OpenStreetMap on each tile’s grid: a 60 m corridor (30 m either side) along each railway, the Konza Techno City plot, and a background of all pixels more than 500 m from any standard-gauge track and outside the plot. The Mombasa–Nairobi line opened in the first year of the record and serves as a control: a method that detected “railway” rather than “new railway” would light it up as strongly as the Naivasha line.

4.5 One-shot similarity search

A query is the reference embedding (Equation 1) of the pixels inside one OpenStreetMap crop field in 2024, and every pixel is scored by cosine similarity to it. The decision threshold τ is calibrated on the query field alone, before any other field is examined, as the 10th percentile of the similarities of the query field’s own pixels; “similar” therefore means “at least as similar as a typical pixel of the starting field”. For tea we use the field around Kericho closest to 2,000 pixels (0.2 km²) and hold out the remaining 51 mapped fields; for rice at the Mwea irrigation scheme, which has only four mapped fields, we use the field closest to 1,000 pixels and hold out three. We report *held-out recall*, the share of held-out field pixels above τ , and the share of the rest of the home tile above τ . Because OpenStreetMap maps only some fields, the latter is an upper bound on false alarms rather than a measurement. We then apply both queries, with the same thresholds, to 33 tiles across 16 kinds of place, from forest and savanna to desert, open water, city and three other farming systems.

4.6 Linear land-cover probe

To test the embeddings as features for a supervised model, we fit a multinomial logistic regression [Pedregosa et al., 2011] (standardised inputs, L2 penalty with $C = 1$, at most 500 iterations) to predict the eight WorldCover classes present in our tiles from the 128 embedding values of 2021, the year of the labels. WorldCover labels are resampled onto each tile’s grid with nearest-neighbour interpolation. We use 24 tiles in 11 regions spanning maize and wheat farmland, tea, irrigated rice, the capital, montane forest, savanna, desert, the Lake Baringo basin, the Konza plain and the Rift escarpment. From each tile we draw up to 1,500 training pixels per class, so rare classes are not swamped, and 50,000 test pixels at random, so test metrics reflect class frequencies. We evaluate with leave-one-region-out validation: train on ten regions and predict the eleventh. Contiguous tiles (the nine Baringo tiles, three Konza tiles and four escarpment tiles) form single regions, so no test tile has a neighbour in training. Because WorldCover has its own errors, we report *agreement* with it rather than accuracy, together with macro-averaged F1.

5 Results

5.1 Coverage and the landmask

Coverage is complete. The TESSERA registry lists between 7,736 and 8,216 tiles per year inside Kenya’s bounding box, but the box also contains parts of five neighbouring countries and the Indian Ocean. Testing each tile centre against the Natural Earth national boundary leaves 4,766 tiles in every year. At about 123 km² per tile this is roughly 591,000 km², against Kenya’s total area of 580,367 km²; the 2% surplus is consistent with tiles that straddle the border and coast. All 4,766 tiles share one identical year list, 2017–2025, so in Kenya the choice of study site is never constrained by data availability (Figure 1). Coverage is even across zones (Table 2), but the country is not: 49% of the

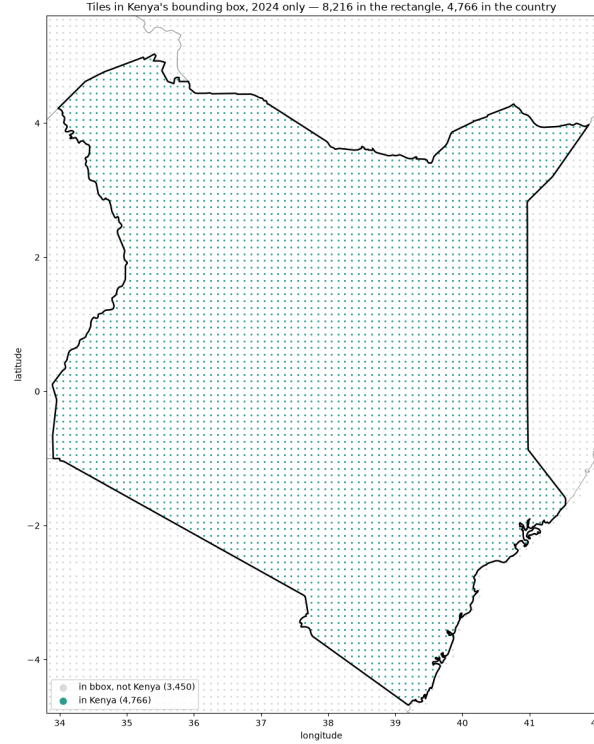


Figure 1: TESSERA coverage over Kenya in 2024. Each dot is one 0.1° tile: 4,766 tiles have their centre inside Kenya (green), and the remainder of the bounding box (grey) belongs to neighbouring countries. Every Kenyan tile has all nine years, 2017–2025.

Table 2: TESSERA tiles over Kenya by zone. Every tile in every zone has all nine years (2017–2025).

Zone	Tiles	Share	Area (km ²)
Arid and semi-arid lands (ASAL)	2,323	49%	287,869
Highlands and Rift	1,574	33%	195,052
Coast	665	14%	82,408
Lake Victoria basin	204	4%	25,280
Total	4,766	100%	590,610

tiles lie in the arid and semi-arid zone and only 4% in the densely farmed Lake Victoria basin, so any uniform sample of tiles is mostly dryland. Clipping by tile centre also drops some coastal tiles that do contain land; Mombasa’s tile, for example, has all nine years in the registry but its centre falls in the Kilindini channel.

The landmask removes the sea but not inland water. Pixels outside the landmask carry 128 exact zeros, which isnan-style checks miss and averages silently include. Table 3 shows the fraction of valid pixels in five probe tiles. Coastal tiles lose 35–40% of their pixels to the sea, while tiles on Lake Turkana and open Lake Victoria are 100% valid. The Turkana tile that scores 100% valid is 96.5% water by Equation 2 (Natural Earth’s lake outline gives 92%), so an average of it taken as a land statistic would mostly describe a lake. In Kenya, “valid” and “land” differ over thousands of square kilometres.

Water embeddings are real signal. That inland water carries embeddings is useful only if those embeddings are meaningful. The water and land references have a cosine similarity of 0.416, whereas the Nairobi reference and a farmland tile near Homa Bay, about 290 km away, have a similarity of 0.744: water is much further from land than two unrelated landscapes are from each other. Applied pixel by pixel to a tile on the western shore of Lake Turkana, with references built hundreds of kilometres away and no spatial smoothing, Equation 2 draws one coherent shoreline (Figure 2), which classification noise could not produce.

Table 3: Share of pixels carrying a non-zero embedding in five probe tiles (2024).

Tile	Setting	Valid pixels
Nairobi	inland control	100.0%
Mombasa island	coast (Kilindini channel)	61.5%
Lamu / Manda	coast (mangrove archipelago)	65.0%
Lake Turkana (Ferguson’s Gulf)	inland lake	100.0%
Lake Victoria	open inland water	100.0%

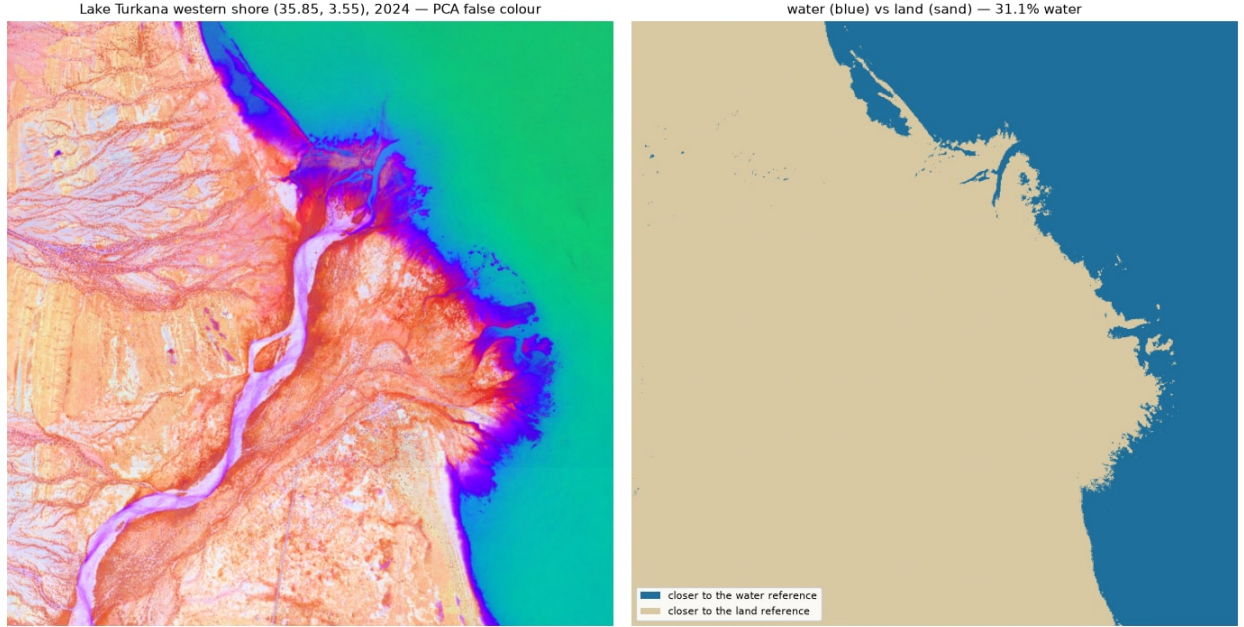


Figure 2: Validation of the two-reference water rule on the western shore of Lake Turkana (tile centred at 35.85°E, 3.55°N, 2024). Left: the embeddings shown as a false-colour composite of their first three principal components. Right: pixels closer to the water reference (blue) or the land reference (sand); 31.1% of the tile is water. The references come from Lake Victoria and Nairobi, and no smoothing is applied.

5.2 Lake Baringo, 2017–2025

The 3×3 block covers 1,112 km² ($3,323 \times 3,345$ pixels of 10 m) and all 81 tile-years are present (about 13 GB). Only 0.10% of the shared grid lacks data, along the edge where the grid overhangs the tiles. As a consistency check, the single tile centred on the lake gives the same water fractions in the mosaic as when processed alone (64.5% in 2017 and 66.6% in 2025).

The lake grew by 42 km². Table 4 and Figure 3 give the lake area by year. It grew from 181.3 km² in 2017 to 223.0 km² in 2025, an increase of 41.7 km² (+23%). The growth was uneven: about 4 km² to 2019, then 33.7 km² in 2020 and 2021, a dip of 7.5 km² in 2023 and a new maximum in 2025. Other water outside the main lake never exceeded 2.4 km², and the lake never touched the edge of the block. Year-by-year maps are given in Appendix A (Figure 9).

Comparison with published estimates. For 2020 we obtain 198.0 km². An independent Landsat-based classification checked with GPS ground points reports 209.3 km² (20,931 ha) [Maina et al., 2026]; our estimate is 5% lower. A government report gives 268 km² [Tobiko, 2021]. The two published figures differ from each other by 28%, which shows how strongly absolute lake area depends on method; our two-class rule probably assigns some shallow, vegetated margins to land, which would bias it low. We therefore regard the size and timing of the change as more reliable than the absolute area.

Table 4: Lake Baringo surface area from TESSERA embeddings. “Other water” is water outside the largest connected component.

Year	Lake (km ²)	Change (km ²)	Other water (km ²)
2017	181.3	—	0.4
2018	184.0	+2.7	0.8
2019	185.7	+1.7	0.6
2020	198.0	+12.4	2.4
2021	219.3	+21.3	0.9
2022	219.1	−0.2	1.0
2023	211.6	−7.5	0.3
2024	211.7	+0.2	1.0
2025	223.0	+11.3	0.8

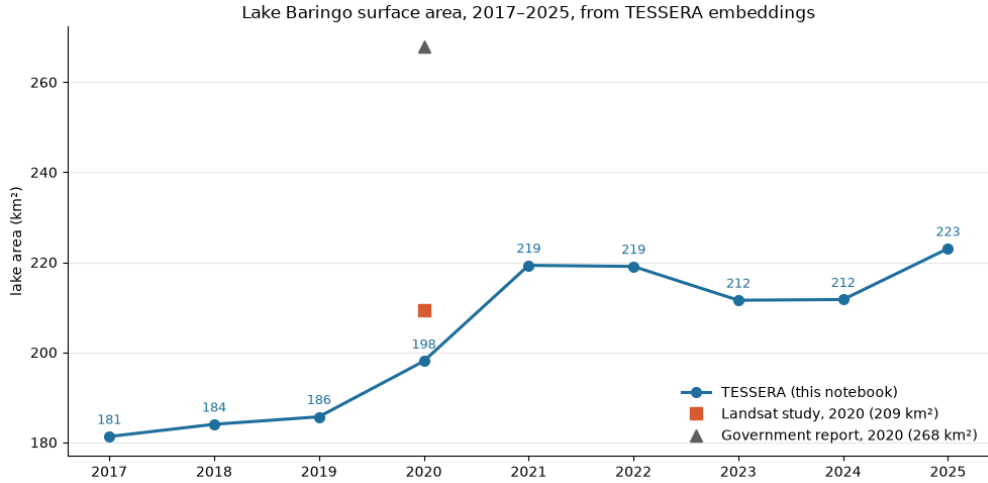


Figure 3: Lake Baringo surface area by year from TESSERA embeddings, with two independent published estimates for 2020: a Landsat-based classification with GPS ground checks (209 km²) [Maina et al., 2026] and a government report (268 km²) [Tobiko, 2021].

Where and when the shoreline moved. Figure 4 colours every pixel that was land in 2017 and lake later by the year it first went under water. The new water forms continuous bands that step outward year by year, widest on the flat southern and eastern shores, rather than scattered pixels. In total 45.9 km² of 2017 land went under water at least once (Table 5); 31.3 km² stayed under water through 2025 and 14.6 km² dried out again. No pixel of the 2017 lake had become land by 2025: the lake only moved outward. Flooding was concentrated in time, with 12.1 km² first flooded in 2020 and 21.3 km² in 2021, together 73% of all newly flooded land.

Temporal stability. Of the 46.4 km² that were lake in some years but not all, 67.4% switched exactly once, from land to lake, and stayed (Table 5). The pixels that switched two or three times follow a small family of sequences with one shape, first under water in 2021 or 2022, dry in 2023 and mostly wet again by 2025, which mirrors the 2023 dip in Table 4. Pixels that switched five or more times, the signature of a noisy classifier, cover about 0.1 km². A core of 180.8 km² was lake in all nine years.

5.3 Beyond water: dating, retrieval and classification

Construction is datable without labels. The 99th percentile of the change magnitude over 785 km² of background is $D = 0.362$. Table 6 reports the results by zone. Along the Nairobi–Naivasha railway 18.4% of corridor pixels changed, 18 times the background rate, and the changes are dated 71% to 2018, 21% to 2019, 8% to 2020 and none later, matching the main construction window of January 2018 to September 2019. Along the control line, which opened in 2017, 2 pixels out of about 9,000 changed, so the method detects new construction rather than railways as such. In the false-colour embeddings the Naivasha line is absent in 2017 and clearly visible in 2025, and the changed pixels follow the mapped track (Figure 5). At Konza the plot changed at 8.6 times the background rate, most often dated 2020 (46%

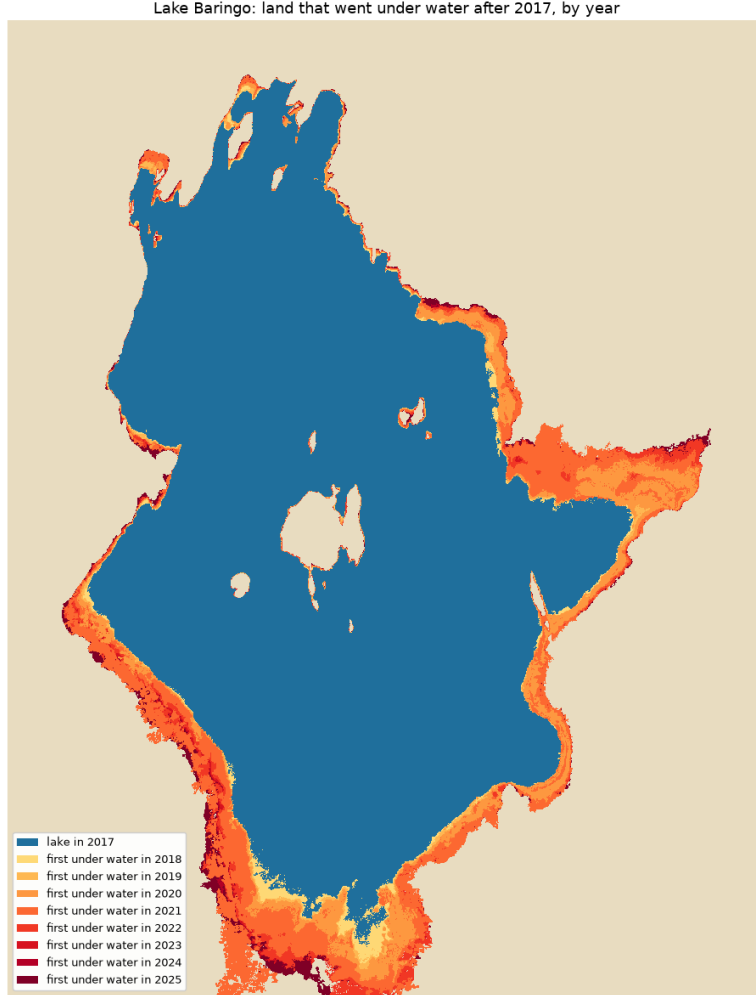


Figure 4: Lake Baringo: the 2017 lake (blue) and land that went under water in later years, coloured by the first year under water. The widest bands are on the southern and eastern shores.

of changed pixels), and the changed pixels form a street grid (Appendix Figure 11). They total 1.50 km² of the 17.4 km² plot, close to the 170 ha (1.7 km²) published for the Phase 1 infrastructure.

Two properties of the dates matter for interpretation. First, even in the background, changes pile up at the ends of the record (32% dated 2018 and 28% dated 2025): a change that happened before 2017, or is still under way, is pushed to the first or last possible year, so dates in the middle of the record are the most precise. Second, only 18% of the Naivasha corridor is flagged; plausible reasons, not tested here, include tunnels and viaducts that leave the surface unchanged, offsets between the mapped and actual track, and a corridor wider than the track formation.

One field finds more of the same crop. Table 7 summarises the retrieval test. The tea query, from a field of 2,067 pixels (0.21 km²), has threshold $\tau = 0.920$ and recovers 59.3% of the pixels of the 51 held-out tea fields (4.87 km²) while flagging 4.0% of the rest of the tile. The rice query (867 pixels) recovers 77.8% of held-out rice pixels; the 20.7% of the Mwea tile flagged outside the mapped fields traces the scheme’s grid of paddies, of which OpenStreetMap maps only four (Figure 6). Across all 33 tiles, the tea query flags 6.35% of the Kericho tea belt on average, 0.03% of the neighbouring Mau Forest tile and 0.00% (to two decimal places) of the other 14 kinds of place, including three other farming systems; the rice query flags only Mwea. This shows the threshold is strict enough not to raise false alarms in different land; it does not show that the query would find tea in another tea-growing region, because none of the other tiles grows tea at scale.

Similarity is also specific. A query from a patch of open water in Lake Baringo matches Baringo’s own water but flags 0.00% of the open-water tiles of Lake Turkana and Lake Victoria, whose median similarities to the query are 0.72

Table 5: Land around Lake Baringo that went under water after 2017, and how often pixels switched between lake and land across the nine annual maps.

Flooded land		Lake/land switches per pixel		
Quantity	km ²	Switches	km ²	Share
Land in 2017, under water at least once	45.9	1	31.3	67.4%
... and under water through 2025	31.3	2	4.6	9.9%
... but dry again by 2025	14.6	3	10.4	22.5%
Lake in 2017, land in 2025	0.0	4	0.0	0.0%
First under water in 2020	12.1	5	0.1	0.1%
First under water in 2021	21.3	6–8	0.0	0.0%

Table 6: Change detection by zone. “Changed” means a change magnitude above the 99th percentile of the background. The right-hand columns give the share of changed pixels whose new state starts in each year; a dash marks zones with fewer than 100 changed pixels.

Zone	Area (km ²)	Changed	Median <i>D</i>	2018	2019	2020	2021	2022	2023	2024	2025
Naivasha line	2.68	18.4%	0.269	71%	21%	8%	0%	0%	0%	0%	0%
Mombasa line (control)	0.90	0.0%	0.201	—	—	—	—	—	—	—	—
Konza plot	17.43	8.6%	0.202	8%	5%	46%	10%	5%	1%	4%	22%
Background	785.02	1.0%	0.186	32%	3%	5%	5%	6%	4%	16%	28%

and 0.45 respectively. Embeddings encode “this kind of place” rather than a category such as “lake”: a single water fingerprint would not find every lake, whereas the relative water-versus-land comparison of Equation 2 works because it only asks which reference is nearer.

A linear probe transfers across regions. In leave-one-region-out validation the logistic-regression probe agrees with WorldCover on 68.6% of held-out pixels, with macro F1 of 0.61 (Table 8). A model trained on all eleven regions and tested on pixels from its own training tiles gives 72.9%, so holding out whole regions costs 4.3 points, a modest gap for a model that cannot memorise places. Agreement is highest where one class dominates (Maasai Mara 94.6%, Konza 89.7%, Mau Forest 88.5%) and lowest in mixed landscapes (Mwea 55.8%, Marsabit 58.1%, Lake Baringo 60.1%). Water is the easiest class (F1 0.88) and wetland the hardest (0.18; 0.3% of test pixels). The confusion matrix (Figure 7) shows the two weak classes are matters of degree: shrubland goes to tree cover 24% of the time, and wetland to cropland 32% of the time, because at Mwea flooded rice paddies are both. On the held-out Kitale maize tile the model reproduces the large fields, the town and tree lines, and disagreement concentrates in the fragmented smallholder patchwork (Figure 8).

The labels, more than the embeddings, limit this result. WorldCover labels Kericho’s tea estates as 64% tree cover and 1% cropland, so a model trained on it cannot map tea as cropland, although the retrieval test above finds tea directly. For Kenyan cropland, WorldCover’s 2020 version has been reported to have a recall of only 0.34 [Kerner et al., 2024], so agreement with it is not evidence of a correct cropland area.

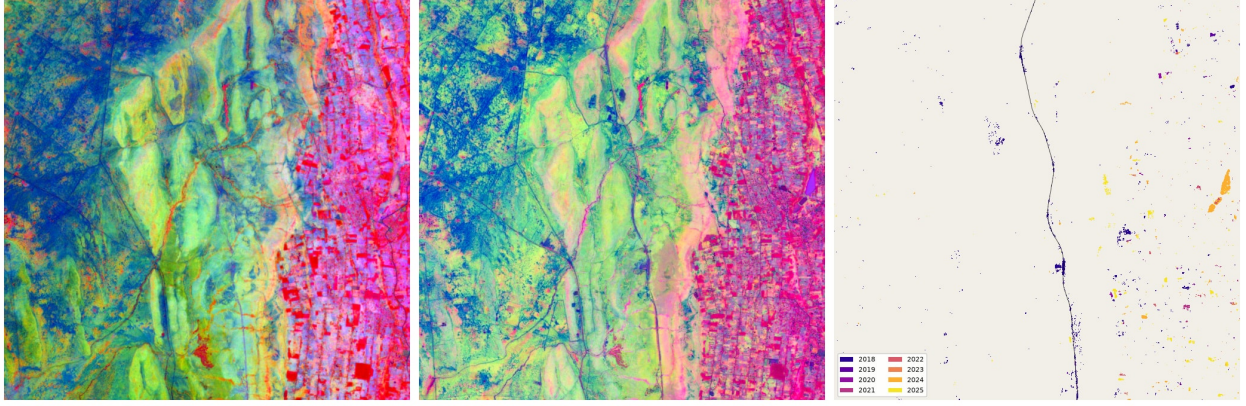


Figure 5: The Nairobi–Naivasha railway on the Rift escarpment (tile centred at 36.55°E , 1.15°S). Left and middle: the embeddings in 2017 and 2025 as principal-component false colour (colours are comparable only within one image). Right: changed pixels coloured by the first year of their new state, with the OpenStreetMap track in black. Changed pixels along the track are almost all dated 2018; most other patches lie in farmland east of the line and are unrelated to the railway.

Table 7: One-shot similarity search seeded with a single OpenStreetMap field. Recall and flag rates are pixel shares. “Rest of tile” is an upper bound on false alarms because many fields are unmapped.

Crop	Query field	τ	Held-out fields	Held-out recall	Rest of tile flagged	Flagged elsewhere
Tea (Kericho)	0.21 km ²	0.920	51 (4.87 km ²)	59.3%	4.0%	$\leq 0.03\%$
Rice (Mwea)	0.09 km ²	0.923	3 (1.16 km ²)	77.8%	20.7%	0.00%



Figure 6: One-shot similarity search. Green: pixels above the threshold calibrated on the query field (red outline). Black: held-out OpenStreetMap fields not shown to the search. Left: tea near Kericho. Right: rice at the Mwea irrigation scheme, where flagged pixels trace the rectangular paddies.

Table 8: Linear probe on TESSERA embeddings, evaluated against ESA WorldCover 2021 with leave-one-region-out validation. Left: agreement for each held-out region. Right: F1 per class over all held-out pixels.

Held-out region	Agreement	Class	F1
Maasai Mara	94.6%	water	0.88
Konza	89.7%	grassland	0.76
Mau Forest	88.5%	tree cover	0.69
Kitale maize	76.1%	cropland	0.67
Nairobi	73.0%	shrubland	0.62
Kericho tea	66.9%	built-up	0.55
Rift escarpment	65.2%	bare / sparse	0.55
Uasin Gishu	63.7%	wetland	0.18
Lake Baringo	60.1%		
Marsabit desert	58.1%		
Mwea rice	55.8%		
All held-out pixels	68.6%	Macro F1	0.61

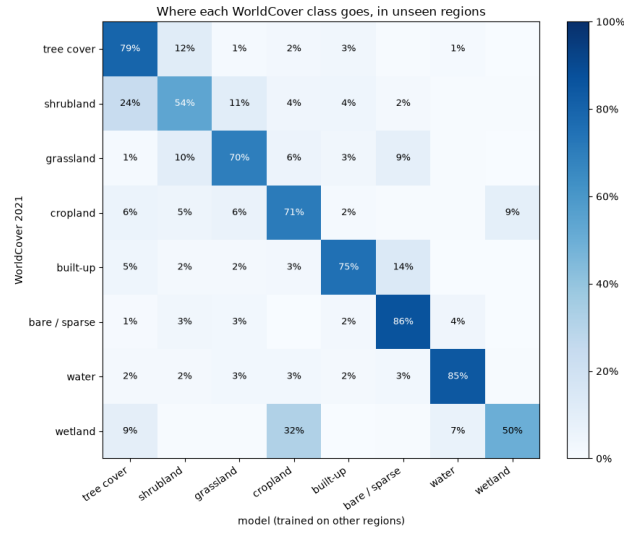


Figure 7: Row-normalised confusion matrix of the linear probe against WorldCover 2021, over all held-out regions. Each row shows where the pixels of one WorldCover class were assigned.

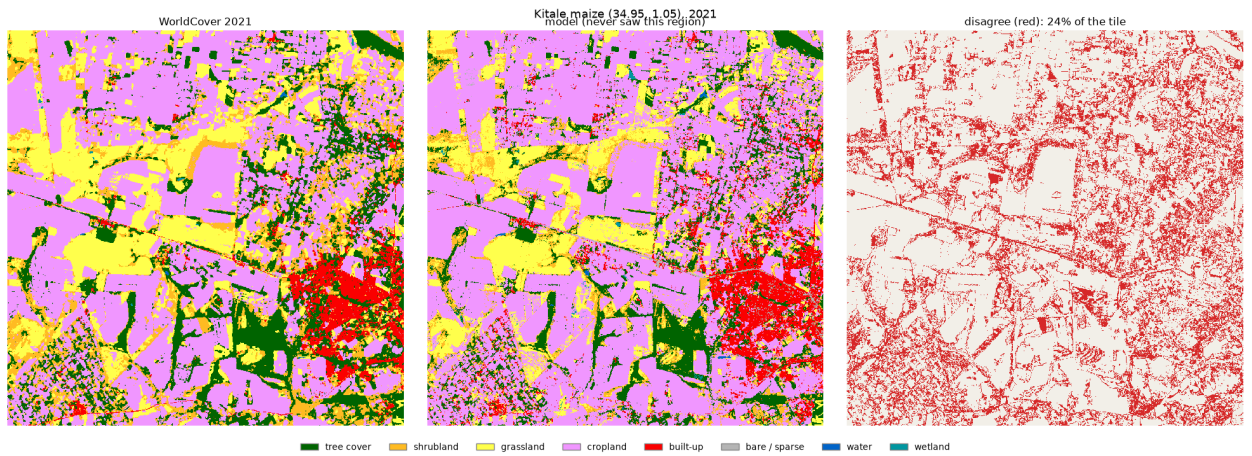


Figure 8: Kitale, a maize region held out from training. Left: WorldCover 2021. Middle: the linear probe trained on the other ten regions. Right: disagreement (24% of the tile), concentrated in the smallholder patchwork.

6 Discussion

What pre-computed embeddings make cheap. Every analysis in this paper reduces to array arithmetic on published vectors: dot products with a few reference vectors, a cumulative sum over years, or one linear model. No model is run, no imagery is cloud-masked or composited, and only the land-cover probe is trained. The nine-year Lake Baringo record, which yields annual extent, the timing of flooding at each pixel and the distinction between permanent and temporary flooding, required two reference vectors and no labelled data. For settings where labelled data are scarce, this changes which questions are affordable.

Validation without ground truth. None of our analyses has dense ground truth, so we rely on three kinds of evidence: agreement with independent estimates (the 2020 lake area), controls that a flawed method would fail (the 2017 railway, tea in non-tea landscapes), and internal consistency (temporal switching, identical results for one tile processed alone and in the mosaic). We recommend designing such checks in from the start when working with embeddings, because a well-formed number is not evidence of a correct one.

Pitfalls. Four details silently corrupt results if missed. (i) Out-of-mask pixels are zero vectors, not missing values, so they enter averages unless filtered; and inland lakes are *inside* the mask, so “valid” does not mean “land”. (ii) Tiles are named by their centres, and in floating point $-1.30/0.1 = -13.000000000000002$, so a naive floor assigns points with negative latitudes to the wrong tile, and roughly half of Kenya’s tiles have negative latitudes. (iii) Change dates are least precise at the ends of the record. (iv) Similarity is specific to a kind of place, so a single fingerprint for a broad category (“lake”) generalises poorly, while relative comparisons between two references are robust.

Cost. One tile-year occupies about 163 MB, so nine years of all Kenyan tiles amount to roughly 7 TB, and the Baringo study alone used 13 GB. In practice only the tiles needed should be downloaded and cached.

7 Limitations

- **Annual resolution.** Each embedding summarises a whole year, so the lake maps describe annual extent rather than seasonal peaks after the rains.
- **Two classes.** Swamp, flooded grass and wet mud are assigned to whichever reference is nearer, which probably explains part of the gap to other lake-area estimates.
- **Record length.** Lake Baringo began rising around 2010 [Tobiko, 2021]; the first large phase precedes the 2017 start of the embeddings.
- **What was lost.** “Land” means “not open water”; whether flooded land held farms, homes or grazing requires additional data.
- **Causes.** The 2023 dip coincides with a multi-season drought in the Horn of Africa, but we do not test the cause.
- **Reference data.** OpenStreetMap maps only some fields, and WorldCover is itself an imperfect product; neither is field-verified ground truth for our tiles.
- **Geographic scope.** The probe uses 24 tiles in 11 regions, with no coastal or mangrove tiles; retrieval has not been tested in a second tea-growing region; and the change threshold was set per study area.
- **No baseline comparison.** We do not compare against spectral-index water mapping or against other embedding products on the same tasks; both are natural next steps.

8 Conclusion

Over Kenya, TESSERA’s pre-computed embeddings provide complete nine-year coverage and support useful measurements with almost no machinery. A comparison against two reference vectors measured the expansion of Lake Baringo from 2017 to 2025, dated when each part of the shoreline went under water, and came within 5% of an independent field-checked estimate for 2020. A single change point per pixel dated the construction of a railway to its documented window while leaving an older control line essentially untouched; a single crop field retrieved more of the same crop with almost no detections in other landscapes; and a linear model transferred across regions. The same approach can be applied directly to Kenya’s other Rift Valley lakes, such as Bogoria, Nakuru and Naivasha, which rose over the same period, and to other monitoring questions where nine years of complete coverage is the point.

Data and code availability

All analysis code, the tile catalogue and the notebooks (K01–K06) are available at <https://github.com/FathirAMM/planetary-ai-tessera-kenya> [Fathir, 2026] under the MIT licence. TESSERA embeddings are released under CC0 and are downloaded on demand with geotessera [University of Cambridge Earth Observation group, 2026a]. Third-party reference data keep their original licences: Natural Earth (public domain), OpenStreetMap (© OpenStreetMap contributors, ODbL 1.0) and ESA WorldCover 2021 (CC BY 4.0; contains modified Copernicus Sentinel data (2021) processed by the ESA WorldCover consortium).

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A Additional figures

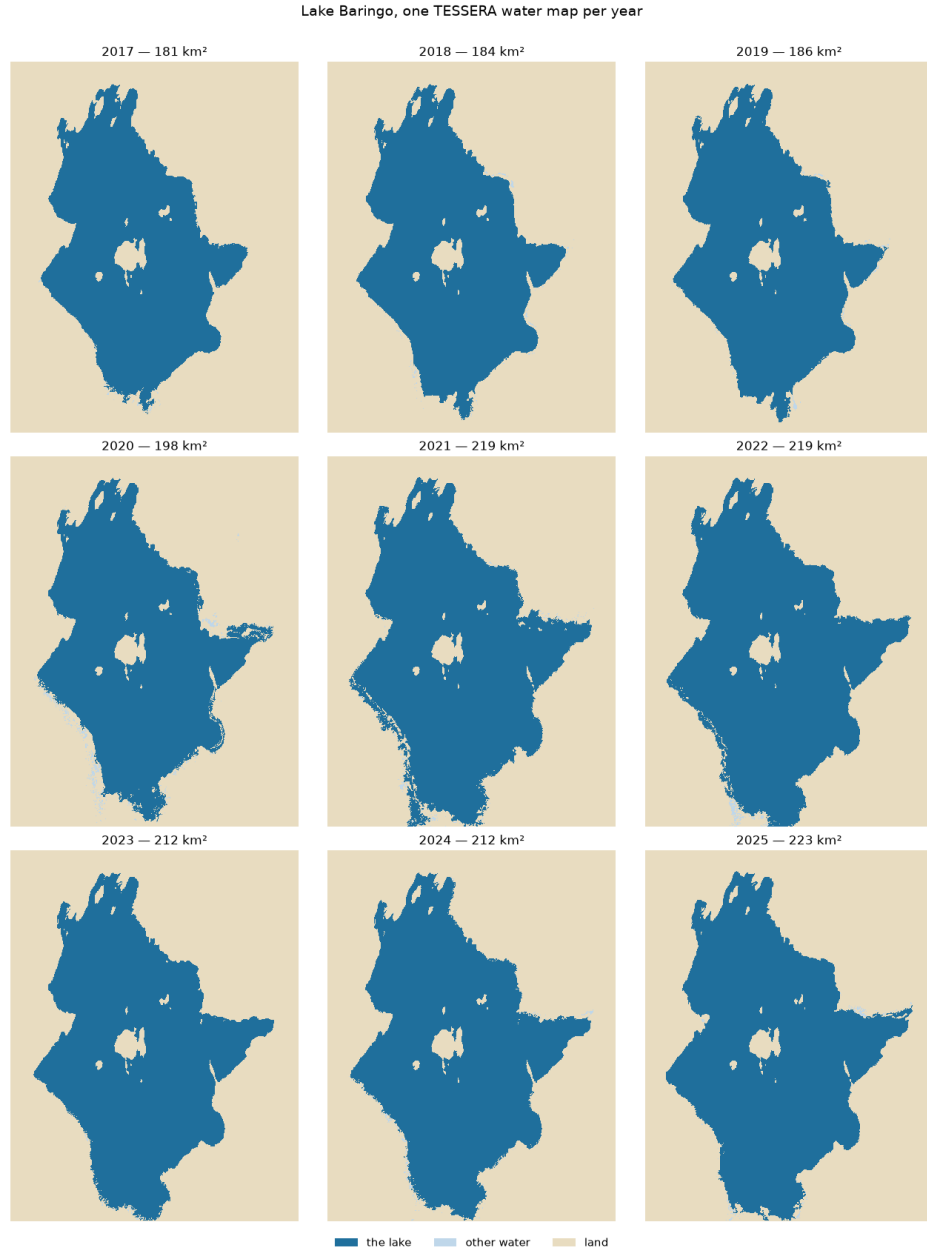


Figure 9: Lake Baringo, one water map per year on the same 10 m grid. Dark blue is the lake (largest connected water component), light blue is other water and beige is land.

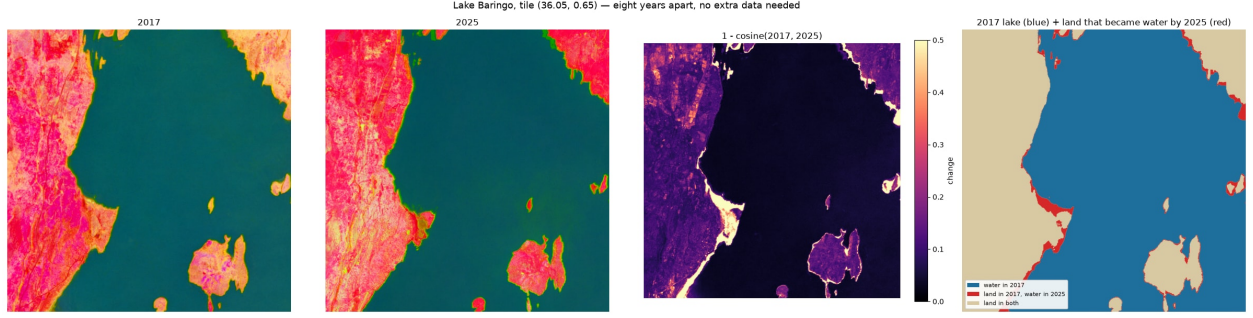


Figure 10: The tile centred on Lake Baringo (36.05°E, 0.65°N). From left: principal-component false colour in 2017 and 2025; change magnitude $1 - \cos(e_{2017}, e_{2025})$; and the 2017 lake (blue) with land that became water by 2025 (red). Pixels that crossed from land to water changed by 0.67 on average, against 0.08 elsewhere.



Figure 11: Konza Techno City: changed pixels coloured by the first year of their new state on two tiles (left: centred at 37.15°E, 1.65°S; right: 37.15°E, 1.75°S), with the OpenStreetMap plot outline in purple. The Phase 1 street grid appears inside the plot, most often dated 2020.

B Minimal example

The water rule of Section 4.2 in full, applied to the tile centred on Lake Baringo. It prints 64.5% water for 2017 and 66.6% for 2025.

```
import numpy as np
from geotessera import GeoTessera

gt = GeoTessera(dataset_version="v1", dataset_variant="vultr")

def tile(lon, lat, year):
    _, emb, crs, transform = next(iter(gt.fetch_embeddings([(year, lon, lat)])))
    return emb

def unit(v):
    n = np.linalg.norm(v, axis=-1, keepdims=True)
    return np.divide(v, n, out=np.zeros_like(v), where=n > 0)

def reference(lon, lat, year=2024):
    return unit(unit(tile(lon, lat, year).reshape(-1, 128)).mean(0))

water = reference(33.95, -1.05) # open Lake Victoria
land = reference(36.85, -1.25) # central Nairobi

for year in (2017, 2025):
    emb = unit(tile(36.05, 0.65, year)) # Lake Baringo
    is_water = (emb @ water) > (emb @ land)
    print(year, f"{is_water.mean():.1%} of the tile is water")
```